

AI FITTRACK

PHASE 6 – PROJECT TESTING

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1. TESTING OVERVIEW

The Project Testing phase verifies the working AI FitTrack backend through server checks, authentication testing, workout CRUD and search testing, AI endpoint testing, error and security checks, and final regression verification. Testing was performed against the supplied Node.js/Express implementation using Thunder Client for manual API verification.

The testing record uses observed results from the project execution. Credentials, passwords, JWT values, API keys, and MongoDB credentials are intentionally excluded from the documentation.

2. TEST ENVIRONMENT

Item	Verified Environment / Tool
Operating Environment	Windows local development environment
Runtime	Node.js v24.19.0
Package Manager	npm 11.17.0
Backend	Node.js + Express.js
Database	MongoDB Atlas
ODM	Mongoose
AI Service	Google Gemini through @google/genai
API Testing	Thunder Client
Development IDE	Visual Studio Code
Server Port	5000

Dependencies were installed successfully using npm. The initial local MongoDB connection attempt returned ECONNREFUSED; MongoDB Atlas was subsequently configured through MONGO_URI and the application connected successfully.

3. SERVER AND HEALTH TESTING

Test	Endpoint / Action	Observed Result	Status
Dependency installation	npm install	Installation completed successfully.	PASS
Server startup	npm run dev	Server running in development mode on port 5000.	PASS
Database connection	MongoDB Atlas	MongoDB connection established successfully.	PASS
Health endpoint	GET /	Success response returned with version 1.0.0.	PASS

```

{
  "success": true,
  "message": "AI FitTrack API is running successfully",
  "version": "1.0.0"
}

```

The health endpoint confirms that the Express application is running before API module testing begins.

4. AUTHENTICATION API TESTING

Test Case	Method / Endpoint	Expected Result	Observed Result
Register user	POST /api/auth/register	User account created and JWT returned.	201 Created — PASS
Login user	POST /api/auth/login	Valid credentials return user information and JWT.	200 OK — PASS
View profile	GET /api/auth/profile	Authenticated profile returned using Bearer JWT.	200 OK — PASS

5. WORKOUT API TESTING

Test Case	Method / Endpoint	Observed Result	Status
Create workout	POST /api/workouts	Workout created successfully.	201 Created — PASS
Get all workouts	GET /api/workouts	Authenticated user's workout data returned.	200 OK — PASS
Search by name	GET /api/workouts/search?q=Morning	Matching workout returned.	200 OK — PASS
Search by category	GET /api/workouts/search?q=Running	Matching workout returned.	200 OK — PASS
Search by date	GET /api/workouts/search?q=2026-09-24	Matching workout returned.	200 OK — PASS
Get by ID	GET /api/workouts/:id	Exact generated ID returned the workout.	200 OK — PASS
Update workout	PUT /api/workouts/:id	Workout values updated successfully.	200 OK — PASS
Delete workout	DELETE /api/workouts/:id	Workout deleted successfully.	200 OK — PASS
Post-delete verification	GET /api/workouts	Empty result confirmed after deletion.	200 OK — PASS

6. AI API TESTING

Test Case	Method / Endpoint	Input Basis	Observed Result	Status
Workout recommendation	POST /api/ai/workout-recommendation	Age, fitness goal, experience	200 OK and generated recommendation after model configuration correction.	PASS
Fitness insights	POST /api/ai/fitness-insights	Total workouts, average duration, total calories burned	200 OK and generated insight.	PASS

7. ERROR HANDLING AND SECURITY TESTING

Test Area	Verification	Observed Behaviour	Status
Missing Authorization	Protected endpoint called without Bearer token	Authentication request rejected.	PASS
Invalid JWT	Protected endpoint called with invalid/failed token	Authentication rejected.	PASS
Registration validation	Required registration fields checked	Invalid registration input is rejected.	PASS
Login validation	Credentials checked before token generation	Invalid credentials are rejected.	PASS
Workout ownership	Individual workout operations checked against authenticated user	User-scoped records are protected.	PASS
Unknown route	Unsupported endpoint requested	Not-found handling is invoked.	PASS
Database startup failure	Database connection error path	Connection failure terminates startup rather than continuing uncertainly.	PASS
Secret handling	Credentials excluded from repository/evidence	Sensitive values are not reproduced in documentation.	PASS

8. TESTING OF DATA VALIDATION

Component	Validation / Rule Checked
User	Name, email and password are required; email is unique/lowercase; password is hashed before persistence.
Workout	workoutName is required; category uses the supported category values; duration has a minimum of 1; caloriesBurned has a minimum of 0.
Authentication	JWT is required for protected endpoints.
Workout access	Workout operations use the authenticated user's identity for user-scoped records.
AI Recommendation	age, fitnessGoal and experience are supplied to the recommendation operation.
AI Insights	totalWorkouts, averageDuration and totalCaloriesBurned are supplied to the insights operation.

9. REGRESSION TESTING

Regression Area	Final Check
Server	Server starts successfully on port 5000.
Database	MongoDB Atlas connection remains successful.
Authentication	Register, login, and profile sequence verified.
Workout	Create, retrieve, search, update, delete and post-delete verification completed.
AI Recommendation	Recommendation endpoint verified after Gemini model configuration correction.
AI Insights	Fitness-insights endpoint verified successfully.
Error Handling	Protected-route and error-handling behaviour checked.
Documentation	Test results are mapped to Skill Wallet evidence requirements.

The regression sequence confirms that the Gemini configuration correction did not replace or remove the existing authentication and workout functionality.

10. TEST RESULT SUMMARY

Testing Area	Result
Environment / Dependencies	PASS
Server Startup	PASS
MongoDB Atlas Connection	PASS
Health Endpoint	PASS
Authentication APIs	PASS
Workout CRUD APIs	PASS
Workout Search	PASS
AI Recommendation	PASS after Gemini model configuration correction
AI Fitness Insights	PASS
Error / Security Checks	PASS
Regression Verification	PASS

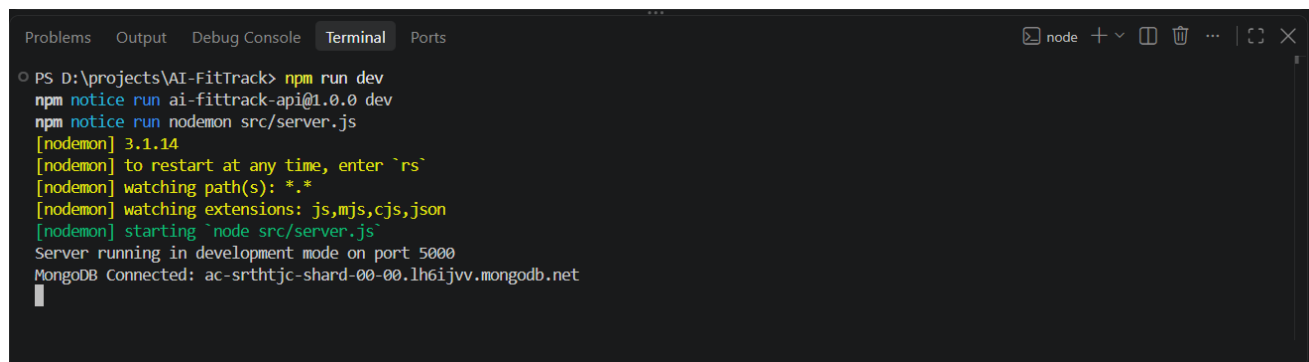
The observed test results cover the major backend functions assigned to AI FitTrack: server configuration, database connectivity, authentication, workout management, search, AI integration, middleware, and centralized error handling.

11. OUTPUT SCREENSHOT EVIDENCE

Evidence No.	Screenshot / Evidence to Store	Skill Wallet Relevance
E1	Server terminal showing successful startup and MongoDB Atlas connection.	Task 14 – Steps for Project Execution / Task 15 – Output Screenshots
E2	GET / health endpoint success response.	Task 15 – Output Screenshots
E3	Register API — 201 Created.	Task 15 – Output Screenshots
E4	Login API — 200 OK.	Task 15 – Output Screenshots
E5	Profile API — 200 OK.	Task 15 – Output Screenshots
E6	Add Workout — 201 Created.	Task 15 – Output Screenshots
E7	Get All Workouts — 200 OK.	Task 15 – Output Screenshots
E8	Search by name/category/date — 200 OK.	Task 15 – Output Screenshots
E9	Get Workout by ID — 200 OK.	Task 15 – Output Screenshots
E10	Update Workout — 200 OK.	Task 15 – Output Screenshots
E11	Delete Workout and post-delete Get All.	Task 15 – Output Screenshots
E12	AI Workout Recommendation — final 200 OK.	Task 15 – Output Screenshots
E13	AI Fitness Insights — final 200 OK.	Task 15 – Output Screenshots

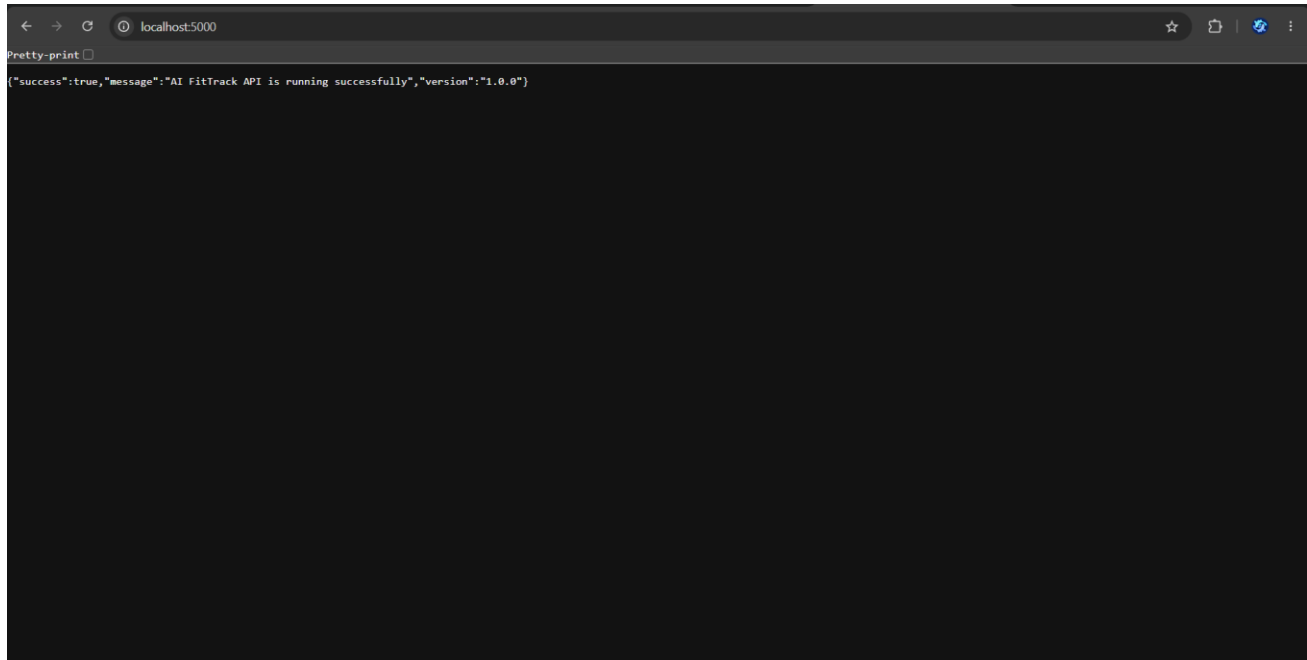
11.1 Server and Database Execution Evidence

Figure 11.1: Server Running and MongoDB Atlas Connection



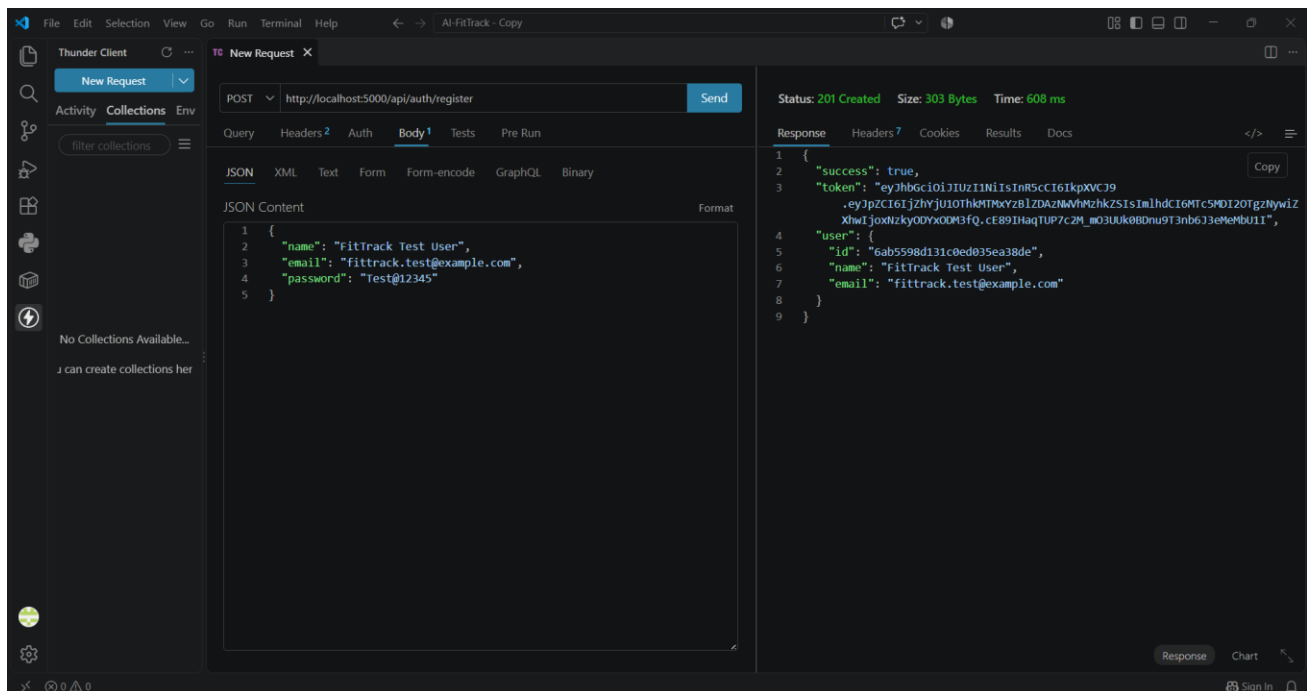
```
PS D:\projects\AI-FitTrack> npm run dev
npm notice run ai-fitrack-api@1.0.0 dev
npm notice run nodemon src/server.js
[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node src/server.js`
Server running in development mode on port 5000
MongoDB Connected: ac-srthtjc-shard-00-00.lh6ijvv.mongodb.net
```

Figure 11.2: Health Endpoint Output



11.2 Authentication API Evidence

Figure 11.3: Register API — 201 Created



[illegible]

The screenshot displays the Thunder Client application, a REST client. The interface is divided into several sections:

- Top Bar:** Contains standard application menus (File, Edit, Selection, View, Go, Run, Terminal, Help) and a breadcrumb path: `Al-FitTrack - Copy`.
- Left Sidebar:**
 - Buttons for `New Request`, `Activity`, `Collections`, and `Env`.
 - A `filter collections` input field.
 - A message: `No Collections Available... you can create collections here`.
- Main Area (Request Editor):**
 - Tabbed interface with `New Request` tabs.
 - Method: `GET`, URL: `http://localhost:5000/api/auth/profile`, and a `Send` button.
 - Query, **Headers** (3), `Auth`, `Body`, `Tests`, and `Pre Run` tabs.
 - HTTP Headers:**
 - ☒ `Accept`: `/*/*`
 - ☒ `User-Agent`: `Thunder Client (https://www.thunderclient.com)`
 - ☒ `Authorization`: `Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.e`
 - ☐ `header`: `value`
 - ☐ `Raw` button.
- Right Panel (Response Details):**
 - Status: `200 OK`, Size: `121 Bytes`, Time: `163 ms`.
 - Tabs: **Response**, `Headers` (7), `Cookies`, `Results`, `Docs`.
 - Response body (JSON):

```

1 {
2   "success": true,
3   "user": {
4     "id": "6ab5598d131c0ed035ea38de",
5     "name": "FitTrack Test User",
6     "email": "fittrack.test@example.com"
7   }
8 }

```
 - Buttons for `Response`, `Chart`, and a copy icon.

11.3 Workout API Evidence

Figure 11.6: Add Workout — 201 Created

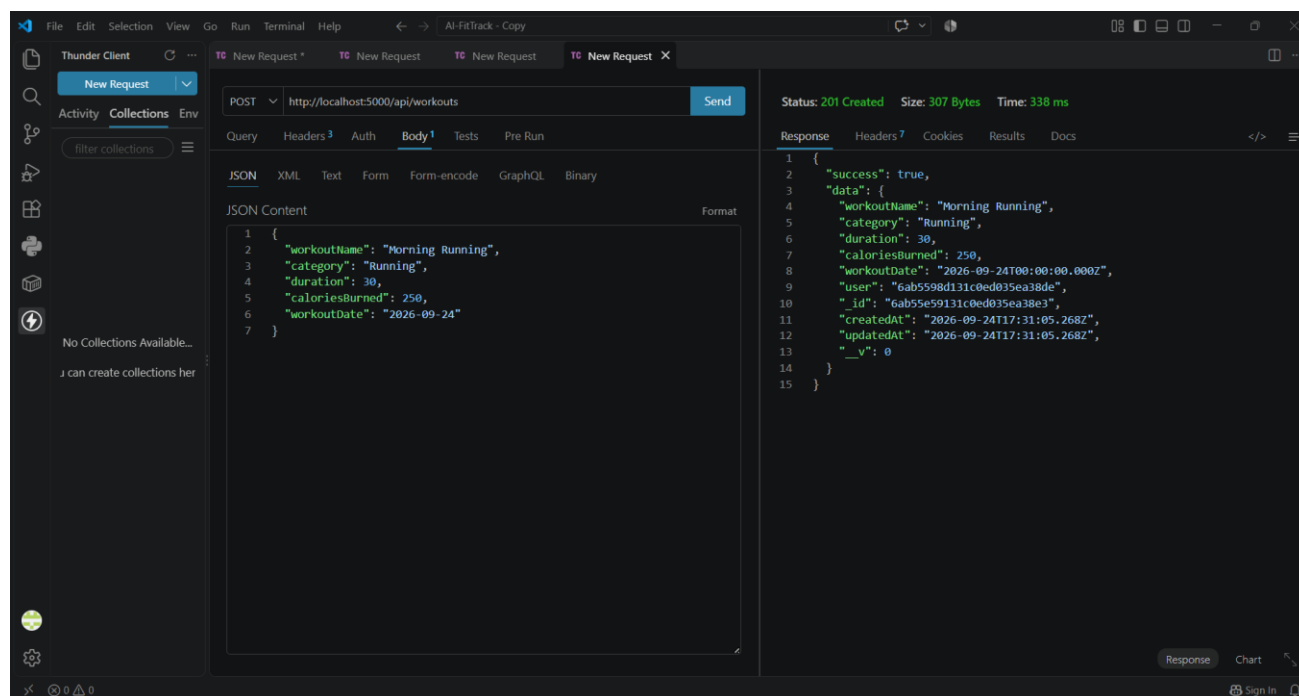


Figure 11.7: Get All Workouts — 200 OK

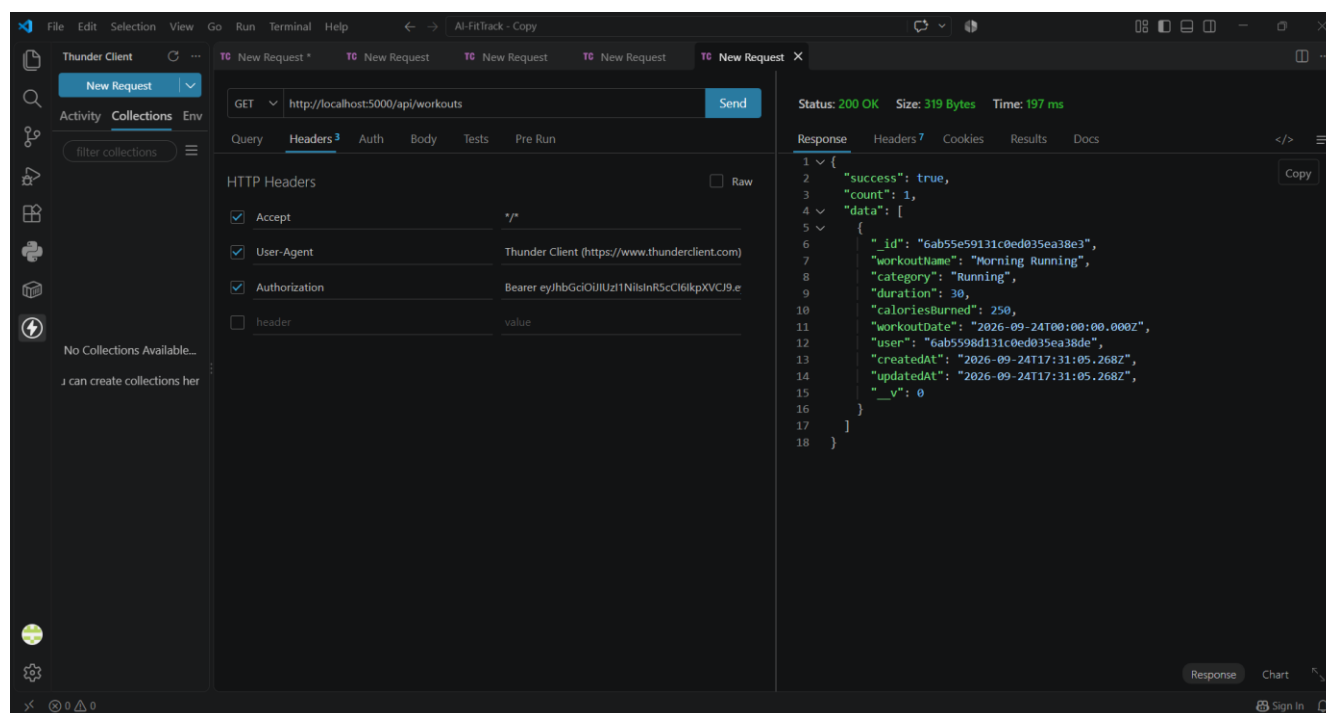


Figure 11.8: Workout Search Results

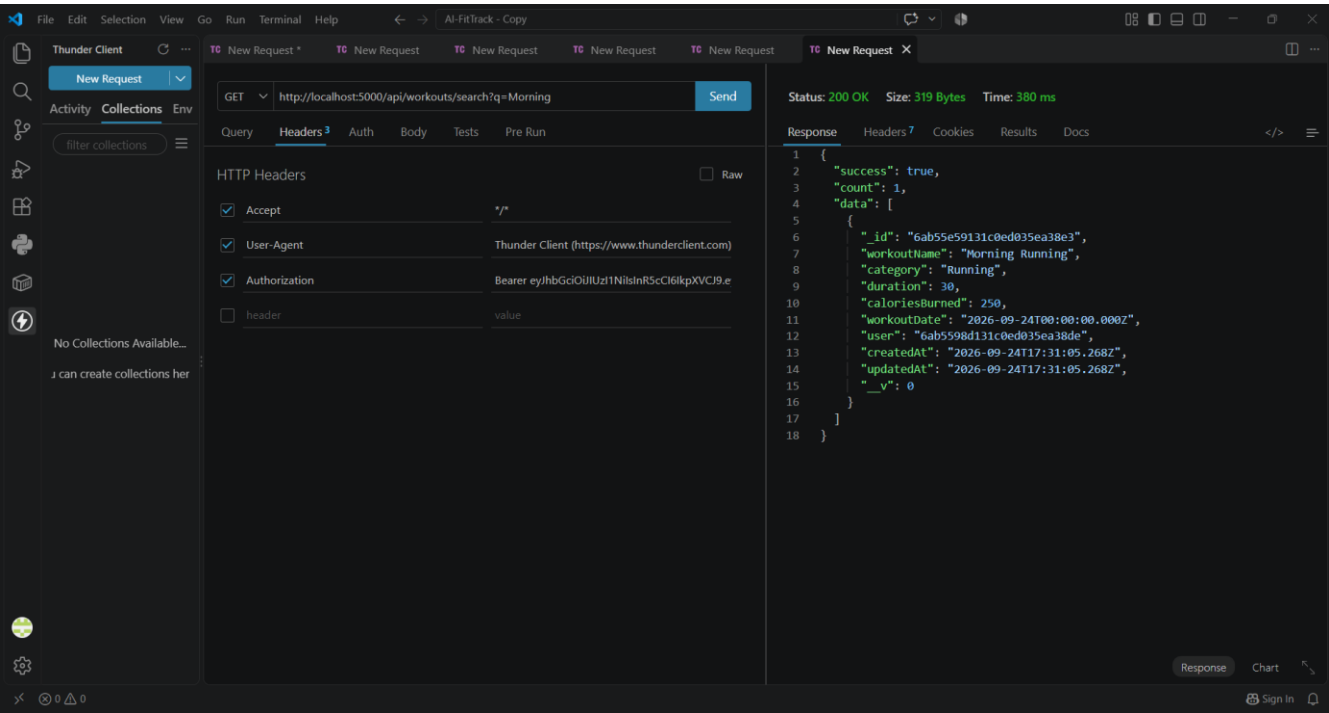


Figure 11.9: Get Workout by ID — 200 OK

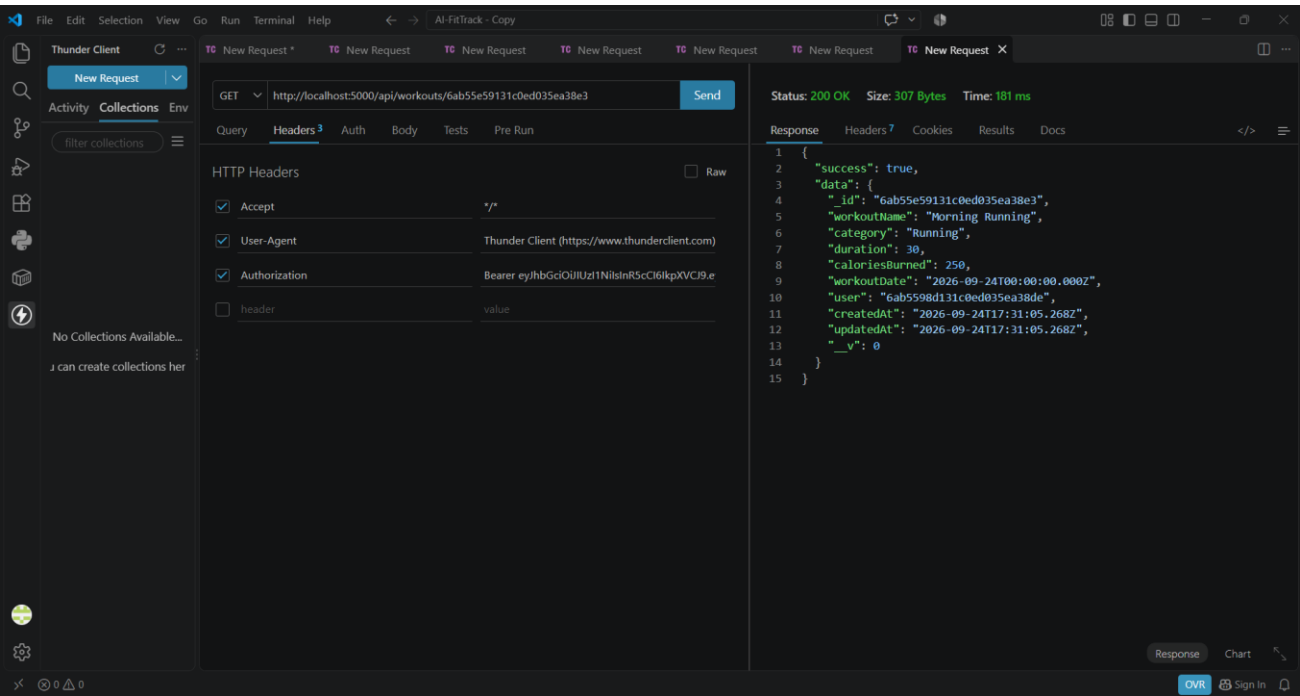


Figure 11.10: Update Workout — 200 OK

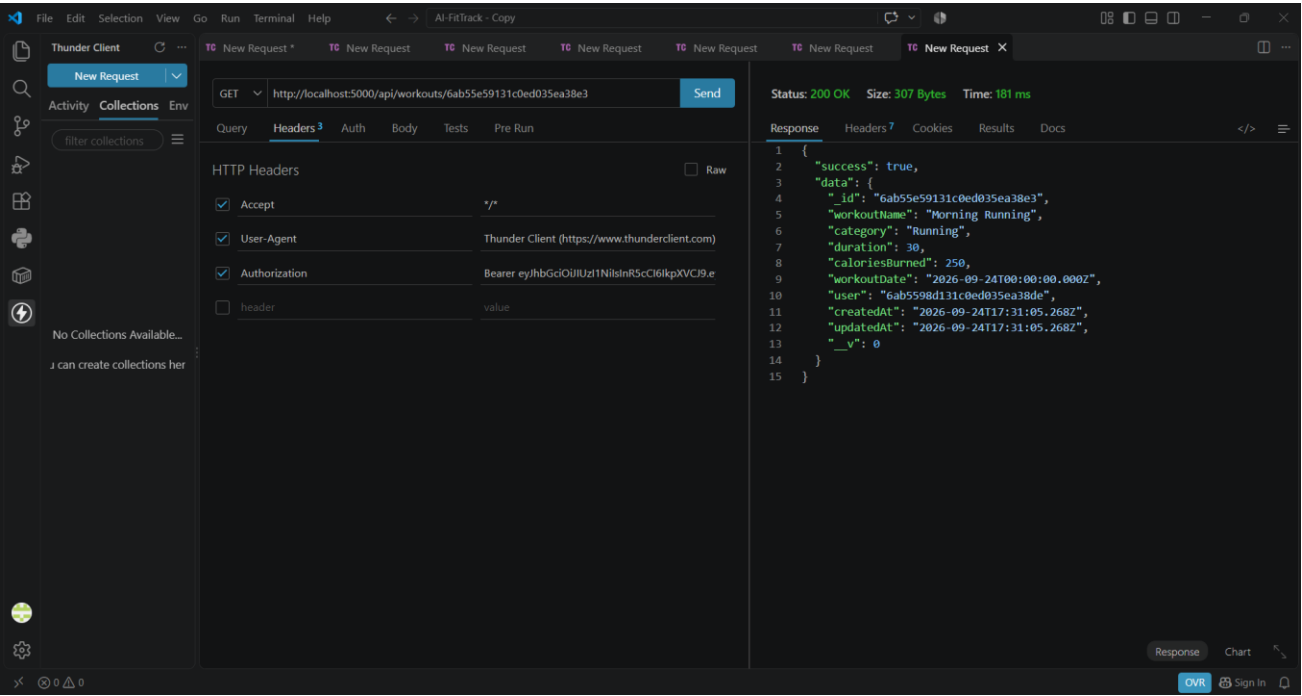
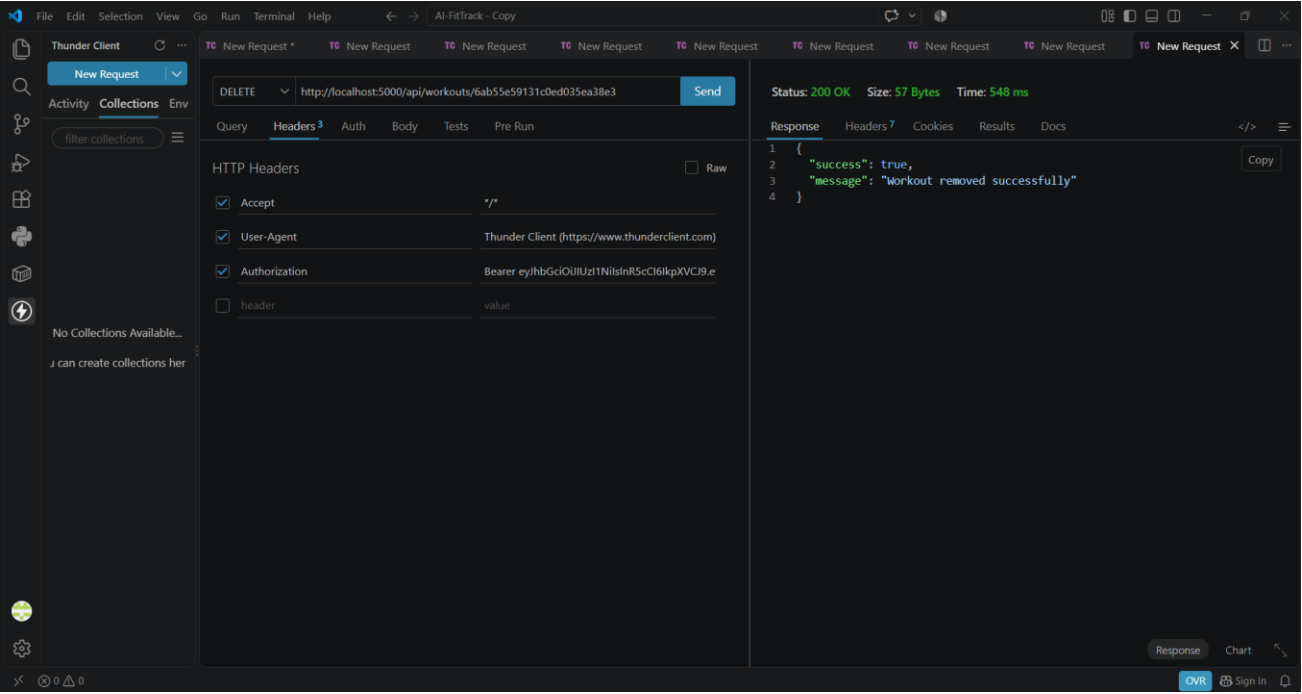


Figure 11.11: Delete Workout and Post-Delete Verification



11.4 AI API Evidence

Figure 11.12: AI Workout Recommendation — 200 OK

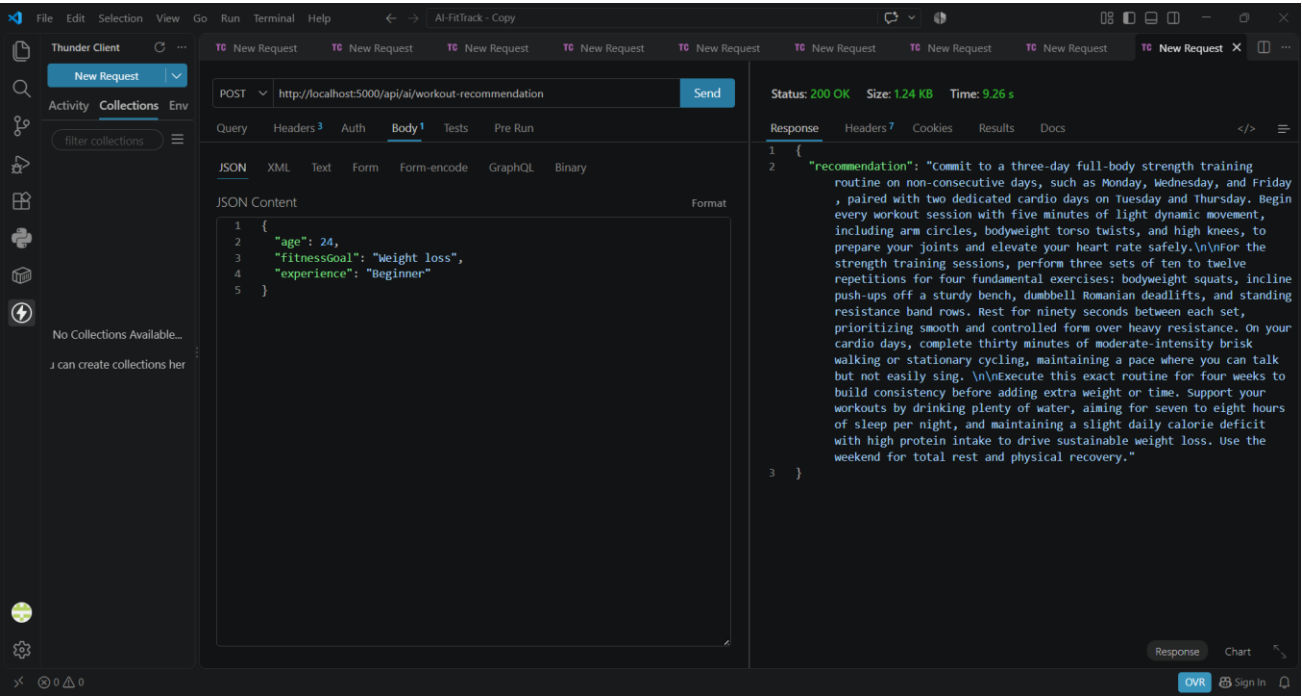
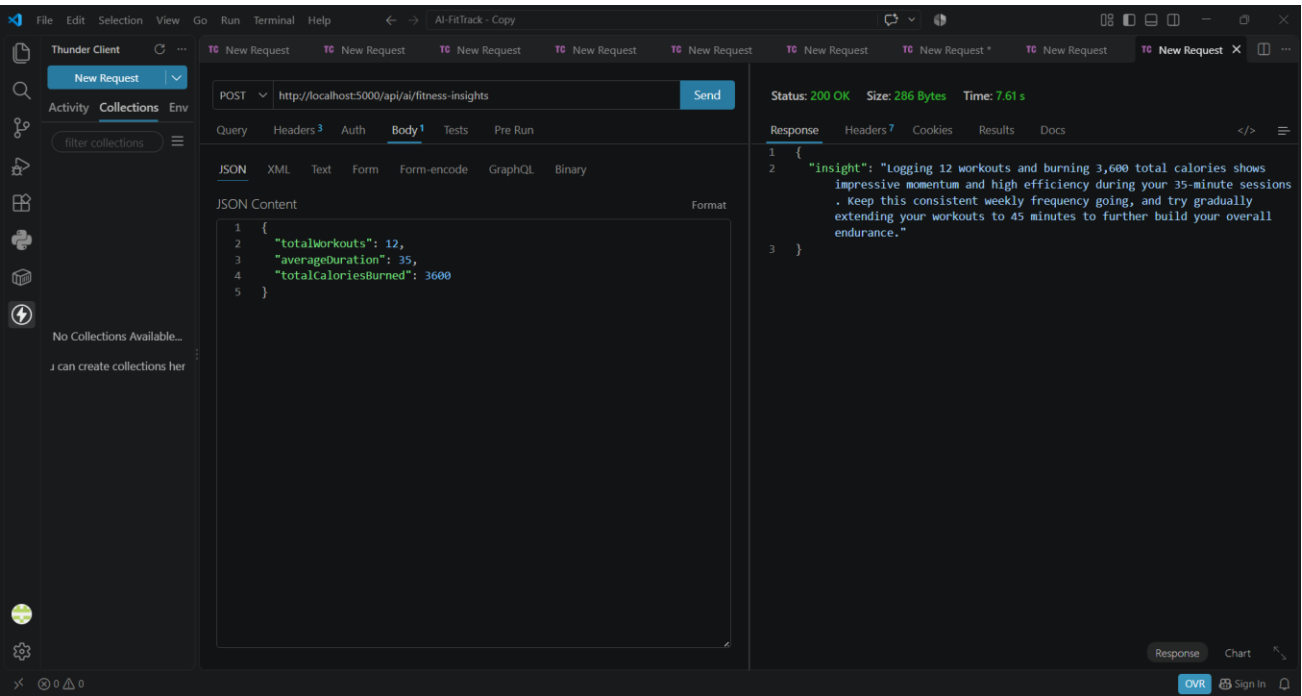


Figure 11.13: AI Fitness Insights — 200 OK



12. TESTING CONCLUSION

The Project Testing phase confirms that the main AI FitTrack backend workflow was exercised from server startup through authentication, workout management, search, AI operations, error handling, and regression verification. The initial Gemini model configuration issue was identified through testing, corrected through the project configuration, and followed by successful AI endpoint verification.